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Simultaneous removal of trace elements from contaminated waters by living *Ulva lactuca*

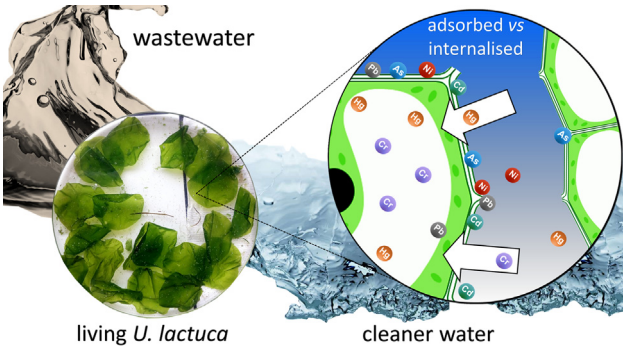
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HIGHLIGHTS

- *Ulva lactuca* was explored to remove As, Cd, Pb, Cu, Cr, Hg, Mn and Ni in wastewater.
- Removal ranged from 48% (As) to 98% (Hg), occurring mostly in the first 12 h.
- Ionic strength (Sal. 15–35) had no influence on macroalgae bioremediation ability.
- Kinetics followed Elovich model, except for Hg where PSO model performed better.
- Hg and Cr were most accumulated, while Ni, Cd and As were retained superficially.

GRAPHICAL ABSTRACT



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ABSTRACT

This work shows the capabilities of living seaweed, *Ulva lactuca*, to remove As, Cd, Pb, Cu, Cr, Hg, Mn and Ni from contaminated waters. Experiments were performed with three algal doses (1.5, 3.0 and 6.0 g L^{-1}, FW), two ionic strengths (salinity 15 and 35), and trace element concentrations corresponding to the maximum allowed values in wastewaters. The highest removals were obtained with the algal dose of 6 g L^{-1}, with efficiencies varying between 48% for As and 98% for Hg, after 24 to 72 h. Salinity showed no effect on the removal efficiency. Overall, Elovich model was the best in describing the kinetics of the process, except for Hg, where pseudo-second-order model performed better. The use of extractions with EDTA (0.001, 0.01 to 0.1 mol L^{-1}) has clarified that most of the Hg (≈98%) and Cr (≈80%) crossed the macroalgae walls, while Ni, Cd and As were retained at the surface (between 60 and 80%). These results support the hypothesis that macroalgae-based technologies may be a viable, cost-effective, and greener option to reduce the rejection of priority hazardous substances in contaminated waters.

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1. Introduction

Sustainable water management in the scenario of global climate change, population growth, and increasing demand of water for agricultural and industrial practices is a key issue (Coppola et al., 2017). One of the main threats to water quality is the presence of metals and metalloids, directly or indirectly released by human activities such as mining, fuel and energy production, use of pesticides and fertilizers, metallurgy and electroplating, leather treatment, batteries or paper industry (Fu and Wang, 2011; Wan Ngah and Hanafiah, 2008; Wang and Chen, 2009).

In particular, arsenic, lead, mercury and cadmium are toxic at very low levels (Cobbina et al., 2015), being classified as priority hazardous substances by the US Agency for Toxic Substances and Disease Registry (ATSDR, 2015). The European Water Framework Directive (WFD) (Directive 2013/39/EU, 2013) foresees strict discharge limits for these elements in wastewaters, which includes total elimination of Hg and Cd by 2021. Classic procedures for the removal of As, Pb, Hg and Cd from wastewaters are ineffective and/or very expensive, which translates into low cost-benefit (Fu and Wang, 2011). In the past decade, innovative technologies based on the sorption of contaminants onto naturally available biomasses (e.g. algae, fungi and bacteria) and agro-industry by-products and wastes (e.g. fruit peels, egg shells, crab shells and cork stoppers) with low economic value have been tested (Abdelfattah et al., 2016; Ahluwalia and Goyal, 2007; Ibrahim, 2011; Jain et al., 2016; Volesky and Naja, 2007). In part, because most of the works were devoted to the removal of a single-contaminant in synthetic water, usually at unrealistic concentrations of contaminant and/or biosorbent, the transition of biosorption-based technologies from laboratory to end-users has being postponed (Michalak et al., 2013; Park et al., 2010). Only few studies addressed real levels of water contamination and multi-contaminant scenarios (e.g. Khan et al., 2017 and Tsekova et al., 2010).

Among biosorbents, algae have shown a notable capacity to retain metallic cations (Cobas et al., 2014; Sari and Tuzen, 2008), due to the presence of diverse negatively charged functional groups in cell walls, such as amine, hydroxyl, carboxylate, phosphate and carbonyl (Costas and López, 2001). Nevertheless, the potential of living macroalgae and their bioaccumulation capabilities have been little explored (Chojnacka, 2010; Henriques et al., 2017a). Studies focusing on the uptake of contaminants by seaweeds usually report only total concentrations, and cellular location of contaminants in biomass is not addressed, which could bring insights into the mechanisms (Costas and López, 2001). In a previous study we demonstrated the capability of living macroalgae in the removal of Hg from mercury contaminated saline waters, with special emphasis on *Ulva lactuca* (Henriques et al., 2015). Its cosmopolitan distribution, withstanding high variations of water salinity, high growth rates, and tolerance to contaminants (Nikolaisen et al., 2011) are a promising set of characteristics for environmental applications (Ben-Ari et al., 2014; Bruhn et al., 2011; Chen et al., 2015). Moreover, it has a large surface area due to its thin and sheet-like thallus, and a relatively simple structure, with uniform and physiologically active cells that facilitates the interactions between the several functional groups in the surface and the metallic cations in solution (Sari and Tuzen, 2008).

Hence, the objective of this work was to examine whether living *U. lactuca* has potentialities in the clean-up of wastewaters contaminated simultaneously by As, Cd, Cr, Cu, Hg, Mn, Ni and Pb. Three doses of algal biomass (1.5, 3.0 and 6.0 g L^{-1}) were tested for the removal of realistic element concentrations from waters of salinity 15 and 35.

2. Material and methods

2.1. Reagents

All reagents used were of analytical grade. HNO_3 65% Suprapur® was purchased from Merck while mono-element standard solutions (1000 mg L^{-1}; in nitric acid matrix) were purchased from Merck (As (V), Cd(II), Cu(II), Hg(II) and Pb(II)), from Inorganic Ventures (Cr(III) and Mn(II)) and from Spectrosol (Ni(II)). All glassware and plastic material were acid-washed prior to use.

2.2. Sampling of Ulva lactuca

Ulva lactuca were collected from the Ria de Aveiro (Portugal, 40°35′12″N, 8°44′55″W) during low tide and transported to laboratory in isothermal plastic bags containing local water. After rinse with clean seawater to remove debris and epibionts, an algal sample was immediately freeze-dried for determining the present concentrations of the inorganic contaminants under study. Remaining seaweed were kept in aquaria for 3 weeks, filled with filtered seawater under natural light (approximately 12L:12D), at room temperature of 20 ± 2 °C. The ratio between dry weight and fresh weight of *U. lactuca* was determined as described in Henriques et al. (2017b). Seawater used for algae cultivation and for the experiments was collected at the coast of Aveiro (Portugal, 40°32′57″N 8°46′30″W), filtered through 0.45 μm pore size filters and stored in the dark at 4 °C until further use. A brief characterization of the seawater, including pH, salinity, conductivity (recorded on a WTW meter) and multi-elemental analysis (performed on a Thermo ICP-MS XSeries) may be found in the Supplementary information (Table S1).

2.3. Exposure of Ulva lactuca to multi-contaminant solutions

Capacity of the living *U. lactuca* to remove As, Cd, Cr, Cu, Hg, Mn, Ni and Pb from contaminated waters was assessed by exposing 1.5, 3.0 and 6.0 g L^{-1} of fresh macroalgae to filtered seawaters (salinity 15 and 35) containing the maximum allowable concentrations of those elements in effluent discharges (Decree-Law No. 236/98, 1998; Directive 83/513/EEC, 1983; Directive 84/156/EEC, 1984): 1 mg L^{-1} for As, Pb and Cu; 2 mg L^{-1} for Cr, Mn and Ni; 0.2 mg L^{-1} for Cd; and 0.05 mg L^{-1} for Hg. Salinity 15 was obtained by pre-diluting filtered seawater with ultrapure water (18.2 MΩ cm). The pH was adjusted to 7.8 with NaOH (1 mol L^{-1}) and all spiked solutions were left to pre-equilibrate during 12 h before the experiments and monitored during the assays. Discs of *U. lactuca* with uniform size (Ø = 58 mm; weight = 0.300 ± 0.04 g) were cut from non-discoloured and healthy thallus and left to acclimatize in clean seawater for 15 h prior the start of experiments. Exposures lasted 72 h, in Schott© glass flasks containing 1 L of solution, under natural light (12 h light:12 h dark) at room temperature (20 ± 2C°). Along the exposures, 5–10 mL of water was sampled at time intervals of 0 h (just before the addition of macroalgae) and at 1, 3, 6, 12, 24, 48 and 72 h. The water samples were immediately acidified to pH < 2 with HNO_3 65% and stored at 4C° until quantification. Experiments were performed in triplicate together with blanks (living macroalgae in clean water) and controls (wastewater in the absence of living macroalgae). At the end of the experiments, macroalgae were harvested for quantification of contaminant concentrations in the biomass.

2.4. Extra-cellular versus intra-cellular element concentrations

Contaminant partitioning between surface (extracellular sorption) and internalized parts (intracellular uptake) of *U. lactuca* was assessed by using extractant solutions (Hassler et al., 2004; Pérez-Llamazares et al., 2010). Sorption and uptake fractions were discriminated through the analysis of element concentrations in the discs of *U. lactuca* with and without pre-washing in ultrapure water, and EDTA 0.1, 0.01 and 0.001 mol L^{-1} in NaCl solution (0.6 mol L^{-1}). Four replicates of the discs were placed into individual Schott flasks, containing 20 mL of washing solution, during 15 min at 150 rpm. The pH of all washing solutions was adjusted to 8 to mimic the pH of seawater.

2.5. Analytical methods

Mercury concentration in water samples was quantified by cold vapour atomic fluorescence spectroscopy (CV-AFS), on a PSA 10.025 Millennium Merlin Hg analyser, using $SnCl_2$ (2% m/v in HCl 10% *v*/v) as reducing agent. In the range of operational conditions used, quantification limit was 0.02 μg L^{-1}, and the acceptable coefficient of variation among replicates was 5% (Henriques et al., 2017a). Mercury content in macroalgae biomass was determined by thermal decomposition atomic absorption spectrometry with gold amalgamation, using a LECO AMA-254, as described by Costley et al. (2000). Analysis was performed directly in the samples (1 to 20 mg, dry weight) without digestion or pre-specific treatment. A maximum coefficient of variation between replicates of 10% was adopted as acceptance criteria. Quantification limit was 0.03 ng Hg (Henriques et al., 2015). Certified Reference Material BCR-060 (*Lagarosiphon major*, 0.34 ± 0.04 μg Hg/g) was used to assess the quality of results. All recovery percentages for Hg were within the range of 92–96%.

Quantification of As, Cd, Cr, Cu, Mn, Ni and Pb in water samples was performed by inductively coupled plasma mass spectrometry (ICP-MS) on a Thermo ICP-MS XSeries equipped with a Burgener nebuliser. The limits of quantification of the method were 1 μg L^{-1} for As, Cr, Ni, 0.2 μg L^{-1} for Mn and Pb, and 0.1 μg L^{-1} for Cd. The acceptable coefficient of variation among replicates was 5%. Contents of As, Cd, Cr, Cu, Mn, Ni and Pb in the macroalgae biomass were quantified by ICP-MS, after microwave-assisted acid digestion. Freeze-dried samples (100–200 mg) were digested in a CEM MARS 5 microwave, firstly with 2 mL HNO_3 (65%) for 15 min at 170 °C, followed by a second microwave cycle with 0.25 mL H_2O_2 (30%) for 15 min at 170 °C. After addition of H_2O_2, the mixture was left to sit for 20 min before the Teflon reaction vessels were tightened and placed in the microwave to allow any vigorous oxidation to vent. Digests were collected into 25 mL polyethylene vessels and the volume made up with ultrapure water. The quality control was assured by running in parallel with the samples procedural blanks (reaction vessels with only HNO_3 and H_2O_2), which were always below the quantification limit, and BCR-60, whose mean recovery percentages were 89 ± 6% for As, 100 ± 5% for Cd, 103 ± 2% for Pb, 92 ± 7% for Cu, 96 ± 7% for Cr, 95 ± 3% for Mn and 100 ± 5% for Ni.

2.6. Data analysis

For a better comparison of data obtained in the assays for the various contaminants, normalized concentrations were used, defined as the ratio between the concentration of contaminant in water at time *t* (C_t), and the initial concentration (C_0), C_t/C_0. The percentage of contaminant removed from the water (*R*, %), at time *t*, was calculated according to the following equation:

$$R = (C_0 - C_t)/C_0 \times 100 \quad (1)$$

The expected concentration of contaminant in macroalgae biomass (q_t, μg g^{-1}) at time *t* was deduced from the mass balance assuming that all contaminant removed from the water was biosorbed or bioaccumulated by the seaweed:

$$q_t = (C_0 - C_t \times V)/m \quad (2)$$

where *V* (L) is the volume of water and *m* (g, dry weight) is the macroalgae mass. The kinetics of contaminant uptake by *U. lactuca* was studied by the adjustment of widely accepted kinetic reaction models (Ho et al., 2000) to the experimental data:

Lagergren pseudo-first-order model

$$q_t = q_e\left(1 - e^{k_1 t}\right) \quad (3)$$

Ho's pseudo-second-order model

$$q_t = \frac{q_e^2 k_2 t}{1 + q_e k_2 t} \quad (4)$$

Elovich model

$$q_t = \frac{1}{\beta}\ \ln(1 + \alpha\beta t) \quad (5)$$

where q_e (μg g^{-1}) is the amount of contaminant bound by unit of mass when equilibrium is attained, i.e. when there is no further variation in the concentration of contaminant in solution; k_1 is the rate constant of pseudo-first-order (h^{-1}); k_2 is the rate constant of pseudo-second-order (g μg^{-1} h^{-1}); α is the initial sorption rate at zero coverage (μg g^{-1} h^{-1}) and β is the desorption constant (g μg^{-1}). Model-fitting was achieved using GraphPad Prism 6 software (Henriques et al., 2015), which was also used to perform all the statistical analysis of results (tests of significance).

The extra-cellular fraction of contaminant (F_{ext}, %), after 72 h of exposure to spiked water, was calculated considering the concentrations of contaminant that were determined in algal discs with ($q_{f\ EDTA}$, μg g^{-1}) and without (q_f, μg g^{-1}) pre-washing with extractant solution:

$$F_{ext} = \left(q_f - q_{f\ EDTA}\right)/q_f \times 100 \quad (6)$$

The fraction of contaminant internalized (F_{int}) was:

$$F_{int} = 100 - F_{ext} \quad (7)$$

3. Results

3.1. The effect of ionic strength on the removal of multi-elements

To test the effect of the ionic strength, the removal of trace elements by *U. lactuca* (1.5 g L^{-1}, FW) was studied at two salinities: 35 and 15. Fig. 1 illustrates the variation of C_t/C_0 for Hg and Pb with the time. Differences in the removal of Hg by the macroalgae between the two salinities were not noteworthy, with an overlapping of most points. This pattern was observed for most of the other contaminants (data not shown). Slight differences were found for Pb, although considering the whole period, there were no statistically significant differences ($\rho > 0.05$) between the two salinities. In the absence of the macroalgae, contaminant concentrations in solution remained stable throughout the assays, revealing that losses due to sorption to glass walls, precipitation or other processes were negligible.

3.2. The effect of algal dosage on the removal of multi-elements

Fig. 2 displays the variation of the simultaneous removal of As, Cd, Cr, Cu, Hg, Mn, Ni and Pb from spiked seawater by 1.5, 3.0 and 6.0 g L^{-1} of living *U. lactuca* along 72 h. Macroalgae-free experiments showed relatively constant concentrations of the elements in solution over time. In the presence of macroalgae, the ratios C_t/C_0 decreased, reflecting the interactions between the contaminants in the seawater and *U. lactuca*. Despite differences among contaminants and the three quantities of algal biomass, most of the kinetic curves are characterized by a fast decline in the first 12 h of the experiment, followed by a less pronounced decrease until 72 h. Among the studied elements, Hg was removed more effectively, with C_t/C_0 ratios varying between 0.15 and 0.02 at the end of experiments. For the other elements, maximum and minimum ratios at the end of assays corresponding to 1.5 and 6.0 g L^{-1} of *U. lactuca*, respectively, were: 0.57 and 0.13 for Pb; 0.58 and 0.26 for Mn; 0.79 and 0.17 for Cu; 0.84 and 0.28 for Cr; 0.77 and 0.23 for Ni; and 0.69 and 0.45 for Cd. Arsenic values decreased almost linearly with the time,

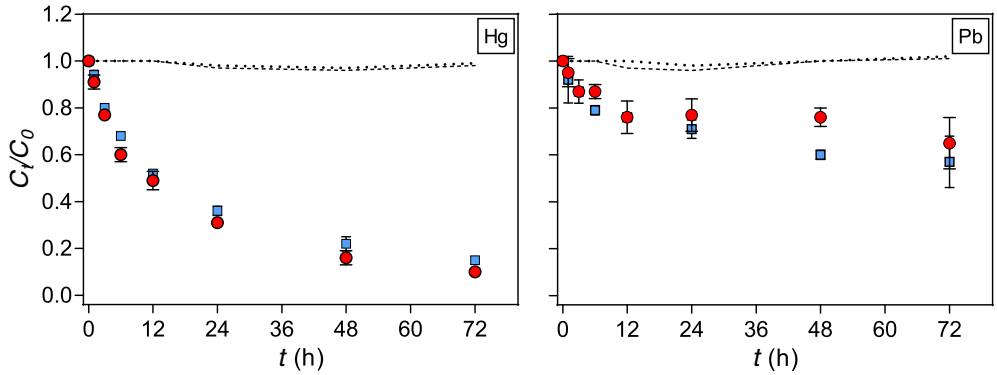


Fig. 1. Concentration of Hg and Pb normalized to the initial value (C_t/C_0) in the contaminated seawater (containing simultaneously As, Pb, Cu, Cr, Mn, Ni, Cd and Hg) at salinity 35 (■) and salinity 15 (●) along time in the presence of *U. lactuca* (1.5 g L^{-1}, FW). Results are expressed as mean values with error bars corresponding to the standard deviation (n = 3). Initial concentrations of 1 mg L^{-1} for Pb and of 0.05 mg L^{-1} for Hg; pH = 7.8.

and after 72 h C_t/C_0 ratios varied between 0.83 and 0.58 for 1.5 and 6.0 g L^{-1} of fresh seaweed, respectively. For Pb, Cu and Cr, differences on removal between the seaweed doses of 1.5 g L^{-1} and 3.0 g L^{-1} were not relevant.

Considering all the analysed elements together, removal increased with the algal dose from 35 ± 22% to 45 ± 23% and 75 ± 16% for 1.5, 3.0 and 6.0 g L^{-1}, respectively. For the highest dose (6.0 g L^{-1}) the removal efficiency followed the order: Hg (98%) > Pb (87%) > Cu (86%) > Ni (77%) > Mn (74%) > Cr (72%) > Cd (56%) > As (48%).

3.3. Kinetic modelling of element removal

The amount of contaminant sorbed or accumulated by living *U. lactuca* (q_t) throughout the exposure to spiked seawater was calculated from Eq. (2) for the sorbent dose of 6.0 g L^{-1}. The q_t values and the adjustments accomplished by pseudo-first-order (PFO), pseudo-second-order (PSO) and Elovich models are presented in the Fig. 3. Results were expressed in dry weight of *U. lactuca*, using a converting factor dry weigh/fresh weigh of 0.25 ± 0.015 (n = 20). This factor is in line with previous values reported by Henriques et al. (2017b).

All the estimated parameters, as well as the goodness of fit, which was assessed by the coefficient of determination (R^2) and the standard deviation of the residuals (Sy.x) are summarized in Table S4 (Supplementary information). The results of kinetic modelling for experiments with 1.5 and 3.0 g L^{-1} of fresh macroalgae are provided as Supplementary information (Table S4 and Figs. S1 and S2). Akaike's Information Criterion (AIC) (El-Khaiary and Malash, 2011) was used for comparison between models, quantifying the probability of the preferential model to be correct. High values of R^2 (0.962–0.999 for 6.0 g L^{-1}) indicate that the three mathematical models used in this study had a good performance in describing the uptake of Hg, Pb, Mn, Cu, Cr and Ni by *U. lactuca*. Agreement between experimental data and models was poorer for Cd and As, with R^2 values ranging between 0.864 and 0.970 (6.0 g L^{-1}). Among kinetic models, PFO achieved the poorest fittings, while Elovich model was the best at describing the experimental results for almost all contaminants and macroalgae/solution ratios (Fig. 3, S1 and S2; Table S4). Nevertheless, PSO showed to be by far the most appropriated model to describe the uptake of Hg by *U. lactuca*, with a probability always higher than 99.99% (AIC). This model accurately fitted the transition between the ascendant part of the curves and the horizontal branch, where PFO and Elovich failed, and predicted quite well the values of pseudo-equilibrium $q_{e,Hg}$ (it was always overestimated by Elovich and underestimated by PFO). Moreover, values of k_2, which relates to rate and extension of sorption, increased with the rise of algal biomass (Table S4, Hg), and were always higher for Hg than for remaining contaminants, agreeing with the behaviour observed in solutions (Fig. 2).

3.4. Trace element concentrations in Ulva lactuca

Table 1 shows the concentrations of As, Cd, Cr, Cu, Hg, Mn, Ni and Pb determined in *U. lactuca* biomass, before and after 72 h of exposure to the spiked seawater. Trace element concentrations in the pristine macroalgae were relatively low, ranging from 0.02 ± 0.007 μg g^{-1} of Hg to 68 ± 18 μg g^{-1} of Mn. Although not shown, concentrations in blanks at 72 h timepoint were statistically indistinguishable from those determined right after field collection. After exposure to contaminated water, *U. lactuca* (1.5 g L^{-1}) showed concentrations of As, Cr, Cu, Mn, Ni and Pb two orders of magnitude higher than the initial values, while those of Cd and of Hg were three and four orders of magnitude greater than the initials, respectively. The contents of Hg, Cd, As, Pb and Mn were higher for the algal dose of 1.5 g L^{-1} than for 3 and 6 g L^{-1}, while Cu, Cr and Ni concentrations were not statistically different ($\rho > 0.05$) among the three algal doses.

The concentrations of As, Cd, Cr, Cu, Hg, Mn, Ni and Pb quantified in *U. lactuca* biomass after 72 h exposure (q_t obtained, μmol/g) were compared to the values expected by the mass balance calculation (q_t expected, μmol/g; Eq. (2)). Fig. 4 illustrates the good agreement between the values, with statistically significant differences (Two-Way ANOVA, $\rho < 0.05$) being only observed for Cr and Ni.

3.5. Trace element partitioning in Ulva lactuca exposed to contaminants

To get more insights into the mechanisms behind the removal of contaminants by *U. lactuca*, an operationally defined partition methodology based on extraction by ligands was followed to discriminate intracellular uptake from surface adsorption. Despite EDTA being considered the best suited washing agent for this type of approach, there is not a consensus in the literature on experimental conditions (see Table S3 in Supplementary information). Hence, ultrapure water, and EDTA 0.1, 0.01 and 0.001 mol L^{-1} in NaCl 0.6 mol L^{-1} were used as washing agents. The percentage of contaminant extracted by the washing or fraction of contaminant bound extracellular (F_{ext}, %; Eq. (6)) is presented in Fig. 5. Ultrapure water extracted small fractions of As, Cd, Cu and Ni, and was not able to extract detectable values of Hg, Pb and Mn. Much higher fractions were obtained with the EDTA solutions, particularly for Cu, Ni, As, Cd and Pb that reached >80% with EDTA 0.1 and 0.01 mol L^{-1}. Best correlations in the EDTA extracts were found for Cr—Mn ($R^2 = 0.99$), Ni—Mn ($R^2 = 0.98$), Cu—Mn ($R^2 = 0.97$) and Pb—Mn ($R^2 = 0.94$). Arsenic and Pb showed poorer correlations with Mn, with $R^2 = 0.64$ and $R^2 = 0.36$, respectively, while no significant

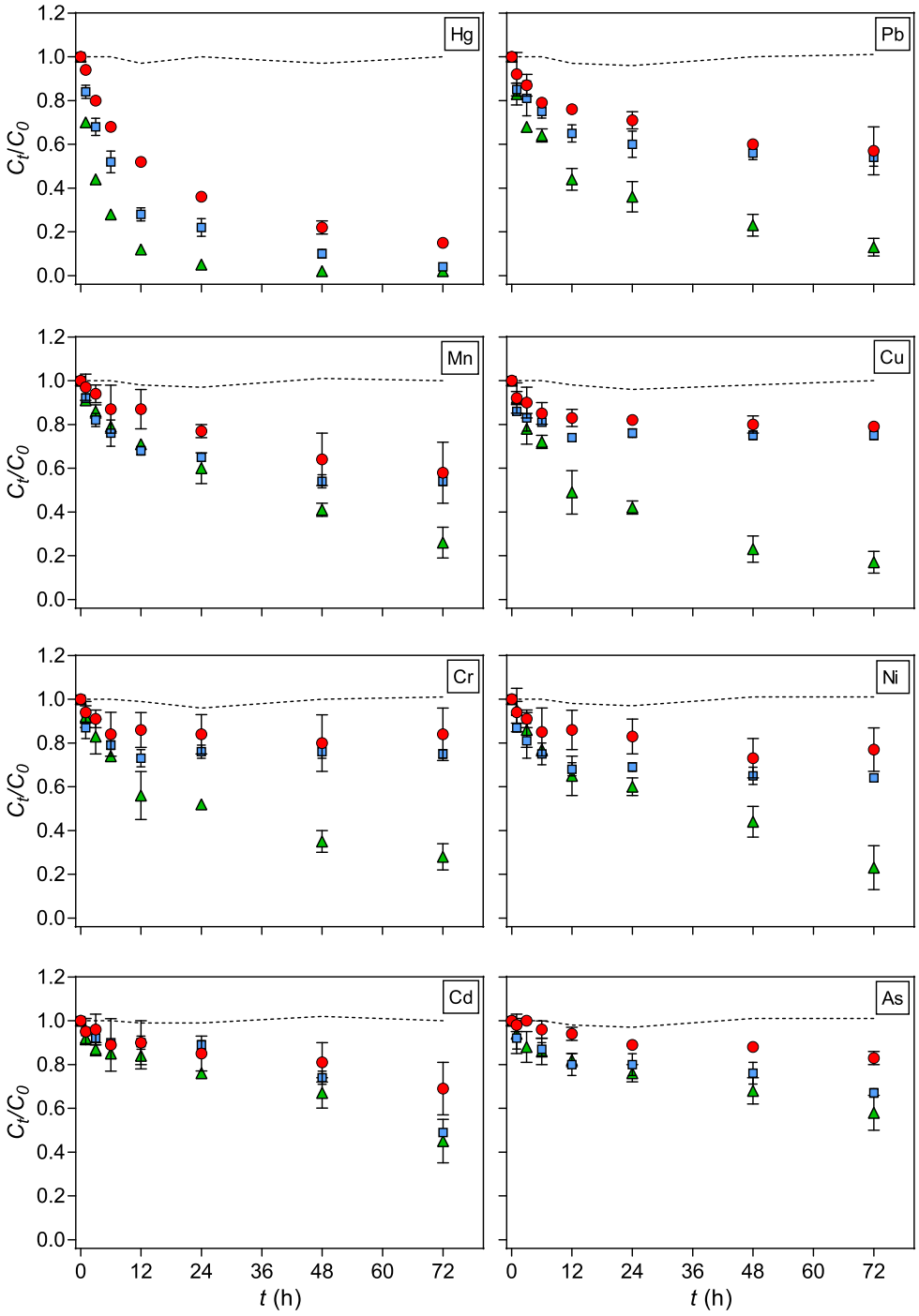


Fig. 2. Concentrations of Hg, Pb, Mn, Cu, Cr, Ni, Cd and As normalized to the initial value (C_t/C_0) in contaminated seawater (salinity 35, pH = 7.8) along time in the presence of *U. lactuca* at the doses of 1.5 (●), 3.0 (■) and 6.0 (▲) g L^{-1} (FW), and in the absence of seaweed (–). Results are expressed as mean values with error bars corresponding to the standard deviation (n = 3). Initial concentrations in the spiked seawater were: 1 mg L^{-1} for As, Pb and Cu; 2 mg L^{-1} for Cr, Mn and Ni; 0.2 mg L^{-1} for Cd; and 0.05 mg L^{-1} for Hg.

relations were found between Cd—Mn and Hg—Mn. Worse correlations were always obtained between trace elements except As and Fe ($R^2 = 0.96$).

Despite all the precautions in the washing procedure it should be admitted that partial rupture of the cell wall may occur. That would lead to the release of contaminants present in the interior. Costas and López (2001) and Pérez-Llamazares et al. (2010) argue that the integrity of the cell wall and membrane can be assessed by quantifying the concentrations of potassium in the extracts, which is mainly located intracellular. The average concentrations of K^+ in *U. lactuca* before washing were 21 ± 2.0 mg g^{-1}, which were higher than values after washing with ultrapure water (3.8 ± 0.57 mg g^{-1}), EDTA 0.001 mol L^{-1} (16 ± 1.6 mg g^{-1}), EDTA 0.01 mol L^{-1} (6.1 ± 1.0 mg g^{-1}), and EDTA 0.1 mol L^{-1} (2.9 ± 0.52 mg g^{-1}). The lowest difference of K^+ concentrations in the macroalgae between before and after washing was obtained for EDTA 0.001 mol L^{-1}, meaning that it is likely to produce the most reliable results. Using the results of this extractant solution, it was possible to obtain the following sequence of trace elements bound more superficially: As, Cd (>60% of the total are sorbed on the surface of the macroalgae), Ni (50%), Pb (40%), Cu (30%), Mn (20%), and non-quantified values of Hg and Cr, which may be interpreted as elements mainly bioaccumulated inside the cells.

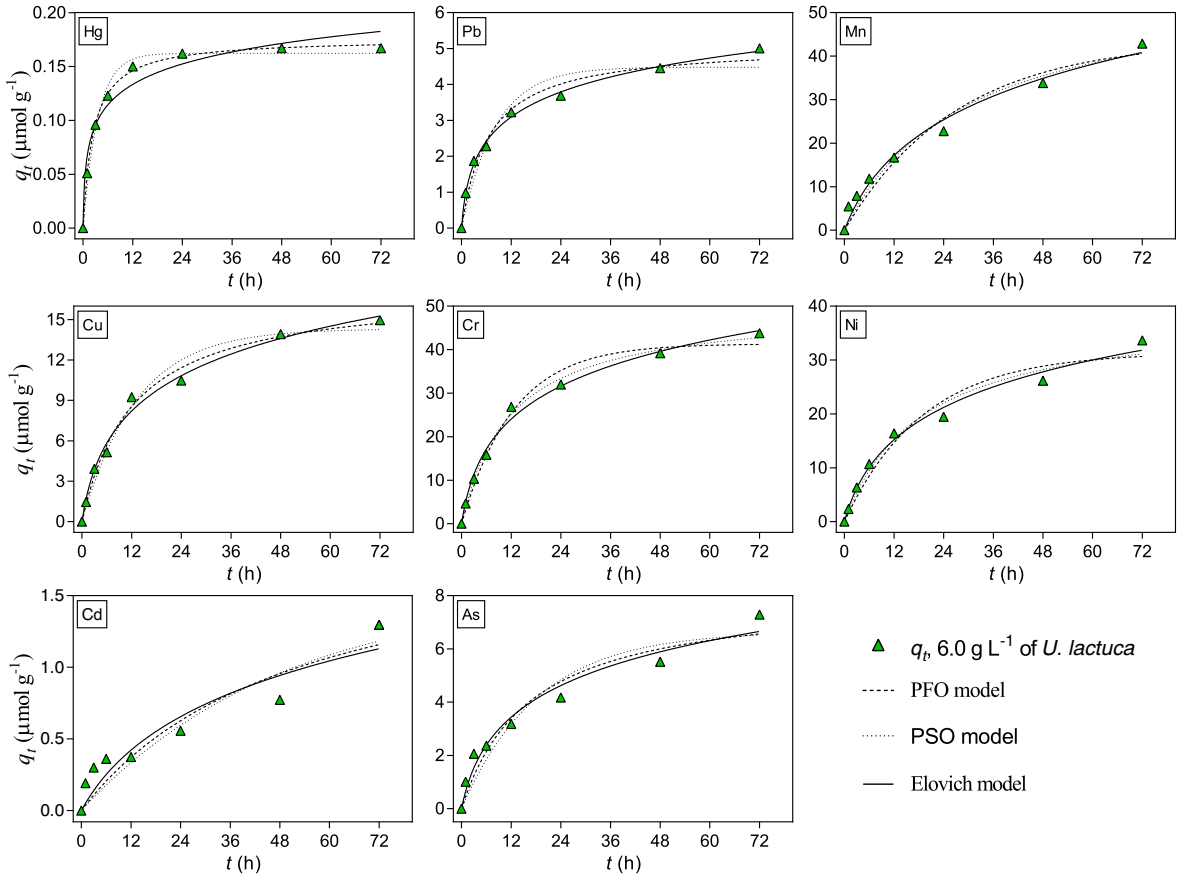


Fig. 3. Kinetic modelling of the experimental data regarding the simultaneous uptake of Hg, Pb, Mn, Cu, Cr, Ni, Cd and As by *U. lactuca*, for an algal dosage of 6.0 g L^{-1} (FW). Error bars were omitted for clarity.

4. Discussion

High accumulation capabilities and tolerance to contamination of marine macroalgae have been already reported (Brinza et al., 2009; Henriques et al., 2015), including for *U. lactuca* (Ben-Ari et al., 2014; Kamala-Kannan et al., 2008; Masakorala et al., 2008; Turner et al., 2008, 2009). However, most studies deal with the removal of a single-contaminant, and the few works on mixtures concern binary or ternary systems. To the best of our knowledge, the present study demonstrated for the first time the capacity of living *U. lactuca* to remove simultaneously several top-priority hazardous substances (As, Pb, Hg and Cd) and other trace elements (Ni, Cr, Cu and Mn) from waters with elements concentrations identical to those that may be found in effluents (limit admissible for wastewaters as reference). The experiments showed that the uptake of trace elements by *U. lactuca* led to a considerable decrease of their levels in water. Removal went up to 98% for Hg with an average of 73% for all the elements, using an algal dose of 6.0 g L^{-1} in a relatively short period of exposure, from 24 to 72 h.

The capacity of *U. lactuca* to remove those elements from water was practically unaffected by salinity between 15 and 35, benefiting from the osmoregulation mechanisms of the macroalgae that enable their growth in a wide range of salinity (Turner et al., 2008). This ability is very promising since wastewaters are rich in salts, which can interfere in the performance of conventional treatment procedures (Adeli et al., 2017; Woodberry et al., 2005). Exposure of the macroalgae to the mixture of 8 contaminants did not cause mortality neither resulted in marked loss of colour or sporulation that are indicative of organism decline (Han and Choi, 2005). The rise in the wastewater pH, from 7.8 to 9.3, identical to that recorded in clean water, both cases in the presence of macroalgae, is closely related to photosynthesis (Cornwall et al., 2012), and supports the absence of severe adverse effects on the organism's metabolism during the experiments.

Table 1
Contaminant concentration in *U. lactuca* biomass (dry weigh basis), before and after 72 h of exposure to spiked seawater (ratio macroalgae biomass/volume solution: 1.5, 3.0 and 6.0 g L^{-1}, FW).

Element	Before exposure	After 72 h of exposure		
		dose 1.5 g L^{-1}	dose 3 g L^{-1}	dose 6 g L^{-1}
	(µg g^{-1})	(µg g^{-1})		
Hg	0.02 ± 0.007	119 ± 7.7	72 ± 9.4	34 ± 3.7
Cd	0.1 ± 0.01	165 ± 12	118 ± 33	72 ± 12
Pb	1.4 ± 0.2	980 ± 112	651 ± 148	446 ± 169
Cr	2.3 ± 0.3	477 ± 98	508 ± 8.3	533 ± 31
Ni	3.1 ± 0.5	664 ± 129	865 ± 52	727 ± 202
As	5.5 ± 0.5	546 ± 125	492 ± 130	298 ± 45
Cu	5.5 ± 0.8	373 ± 51	366 ± 48	373 ± 112
Mn	68 ± 18	1966 ± 420	1428 ± 172	1065 ± 102

4.1. Trace elements removal efficiency

The increase of biomass translates into a larger surface area and hence to high number of binding sites for the contaminants (Kumar et al., 2016). However, a high ratio biomass:water may inhibit growth and survival of the macroalgae (Nikolaisen et al., 2011). Optimising this ratio is also pertinent to minimize the final solid residue, which needs itself proper treatment or storage (Figueira et al., 2016). Ibrahim (2011) investigated the optimum dosage of red seaweed biomass in the biosorption of Co(II), Cd(II), Cr(III) and Pb(II) ions from aqueous solution and concluded that biosorption capacity increases from 1 to 10 g L^{-1} (DW), while for higher doses it occurs partial aggregation of biomass reducing the effective surface area. Sari and Tuzen

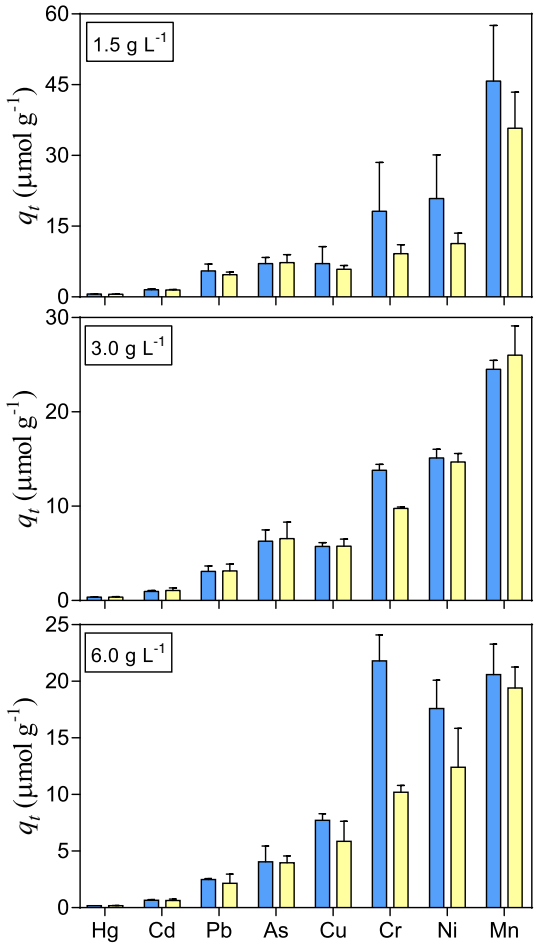


Fig. 4. Expected (■, µmol/g, DW) and obtained (■, µmol/g, DW) concentrations of Hg, As, Cd, Pb, Cu, Cr, Mn and Ni in *U. lactuca*, after 72 h of exposure to contaminated water. Results are expressed as mean values with error bars corresponding to the standard deviation (n = 3).

(2008) studied the effect of *U. lactuca* biomass on the sorption of Pb(II) and Cd(II) and found that efficiency increases with biomass rise from 2 to 20 g L^{-1} (DW), and remain unchangeable for higher doses. The present study tested the removal with lower biomass dosages, 1.5 to 6.0 g L^{-1}, FW, corresponding to 0.4 to 1.7 g L^{-1} in DW.

The use of 3.0 g L^{-1} (FW) brought no relevant advantages relatively to 1.5 g L^{-1}, but the number of binding sites at 6.0 g L^{-1} appears to displace the equilibrium between solution and biomass towards higher removals, especially for Cu, Cr, Ni and Pb. Differences between elements may be related to their speciation in the solution that influences

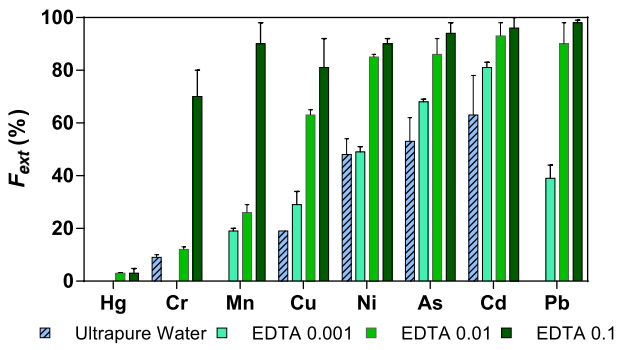


Fig. 5. Fraction of contaminant bound extracellularly (F_{ext}, %) in *U. lactuca*, after 72 h exposure to wastewater, operationally defined by extraction with ultrapure water, and EDTA 0.001, 0.01 and 0.1 M in NaCl 0.6 M.

mobility and affinity towards the macroalgae surface. For example, simulations performed in Visual MINTEQ, using physical-chemical characterization of the water as input parameters indicate that chlorocomplexes of Hg with negative charge dominates (21% $HgCl_3^-$, 75% $HgCl_4^{-2}$), whereas As exists almost exclusively in the form of hydrogen arsenate (96% $HAsO_4^{-2}$). It should be noticed that the initial concentrations in the present study differed among the contaminants, since the main aim was to mimic an industrial effluent and hence maximum allowable values for discharge were used for each element. Consequently, despite the high percentage of Hg removed, the quantity of As (≈6.5 µmol) removed from the solution was higher than that of Hg (≈0.24 µmol).

Apart from adsorption on surface and internalization in the macroalgae biomass, various mechanisms may contribute to the decrease of contaminants in water, such as microprecipitation or colloids formation, or particularly in the case of mercury, volatilization (Kumar et al., 2016).

4.2. Trace element extra- and intra-cellular partition

The rapid initial decrease in contaminant concentrations in solution (Fig. 2) most likely reflects sorption at the cell surface and eventually diffusion in the intercellular spaces (Desaunay and Martins, 2014). This uptake step should be essentially passive (independent of metabolism) and physicochemical. Since the macroalgae surface at the beginning of experiments contains negligible concentrations of contaminants (Henriques et al., 2015), a high driving force exists, which translates into great initial sorption rates. Then, external binding and inward transport by specific proteins will simultaneously occur (Blaby-Haas and Merchant, 2012), as well as internal accumulation of contaminants in the cytoplasm, in cell vacuoles or by attachment to the inner layer of the cell membrane (metabolism dependent) (Costas and López, 2001; Hassler et al., 2004). Kinetic modelling analysis of the present data suggests that more than one mechanism is involved in the uptake process as PFO achieved the worse adjustments to experimental data (Chojnacka, 2010), and that chemisorption in algae surface will be the rate limiting step for most contaminants (Elovich model) (Henriques et al., 2015).

Using data from the washing procedure with EDTA 0.001 mol L^{-1}, which was assumed the extractant solution that altered less the membrane permeability, three groups of trace elements were identified: Ni, As and Cd that are mostly sorbed on the macroalgae surface; Hg and Cr that are mainly bioaccumulated; and Cu, Mn, and Pb that are distributed between the two fractions.

In literature, it is well documented the strong affinity of metals with iron and manganese oxides (Turner et al., 2004), which have an important role in the cycling and transport of metals in aquatic systems. Since the medium, especially near the surface of the macroalgae is rich in O_2, released at photosynthesis, formation of iron and manganese oxides are plausible, as well as their association with the contaminants in solution. Thus, the good correlation of metals to Mn suggests that the fraction of extracellularly bound metal may be associated to Mn oxides in detriment of Fe oxides.

4.3. Possibilities for water reuse and recovery of elements

After 72 h of removal by 6.0 g L^{-1} of *U. lactuca*, Hg in solution was approximately 1 µg L^{-1}, which corresponds to the maximum allowable value in drinking water (Council Directive 98/83/EC, 1998). Quality for human consumption was not achieved for the other trace elements although concentrations of Cu (145 µg L^{-1}), Pb (133 µg L^{-1}) and Ni (500 µg L^{-1}) were in accordance with Portuguese legal criteria for irrigation water (Decree-Law No. 236/98, 1998). Final concentrations of Cd (89 µg L^{-1}) and Mn (454 µg L^{-1}) almost doubled the maximum values for water irrigation (50 and 200 µg L^{-1} respectively). These results highlight the possibility of water reuse for different purposes.

The bioconcentration factor (BCF), defined as the ratio between the contaminant concentration in or on algal biomass after exposure and the initial concentration of contaminant in solution, varied as follows: Hg > Mn > Pb > Cd > Ni > As > Cr > Cu, for 1.5 g of fresh algae per litter of solution, and ranged between 451 of Cu and 2450 of Hg, with an average of circa 1000. These findings point to the promising applicability of *U. lactuca* in pre-concentration and recovery of elements with high strategic or economic value from wastewaters (Jacinto et al., 2017), such as Cr, Mn and Ni (Hennebel et al., 2013). BCF for Hg (dose of 1.5 g L^{-1} of *U. lactuca*) was in accordance with the values reported previously for two different contamination scenarios (Henriques et al., 2017b).

5. Conclusions

This work points out that methodologies based on live *Ulva lactuca* may be a viable option to reduce the rejection of As, Cd, Pb, Cu, Cr, Hg, Mn and Ni in contaminated waters. Results demonstrated that ionic strength (salinity between 15 and 35) has no influence on the macroalgae bioremediation ability. Using an algal dose of 6.0 g L^{-1} (FW), efficiencies varied between 48% for As and 98% for Hg. Almost 50% of removal occurred within the first 12 h. Overall, the model that best fit the experimental data was that of Elovich, except for Hg, whose biosorption/bioaccumulation over time was best described by PSO model. The extraction with 0.001 mol L^{-1} EDTA suggests that As, Cd and Ni were mostly on the surface of the macroalgae, while Hg and Cr was within the algae.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2018.10.282.

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